

REST & SOAP APIs – Complete Beginner to Professional Guide

1. What is an API (Simple Explanation)

An API (Application Programming Interface) is a bridge that allows two different software systems to talk to each other.

For example, a mobile app talks to a backend server to get user data. This communication happens using APIs.

2. What is REST (In Simple Words)

REST is a way of designing APIs so that communication between systems is simple, predictable, and scalable.

REST does not mean a specific technology. It is a style or approach that tells us HOW to design APIs properly.

3. What Does REST Mean (Word by Word)

REST stands for Representational State Transfer.

Representation:

Data is shared in formats like JSON or XML instead of exposing database structures.

State:

State is the current data stored on the server, such as user details or event status.

Transfer:

The representation of state is transferred from server to client using HTTP.

4. REST Definition (Formal)

REST is an architectural style for building APIs that use HTTP methods to operate on resources in a stateless manner by transferring representations of resource state.

5. Core Concept of REST – Resources

In REST, everything is treated as a resource.

Examples of resources:

- User
- Event
- Order
- Product

Each resource is identified using a URL.

6. REST Rules (Principles)

1. Resource-Based URLs (No Actions in URL)

Correct: /users, /orders/10

Incorrect: /getUsers, /deleteOrder

2. HTTP Methods Define Actions

GET – Read

POST – Create

PUT – Update full

PATCH – Update partial

DELETE – Delete

3. Path Variables for Identity

Example: GET /users/10

4. Query Parameters for Search, Filter & Sort

Example: GET /events?status=ACTIVE&type=RFP

5. Request Body for Resource State

Used when creating or updating resources.

6. Stateless Communication

Each request must contain all required data. Server does not remember previous requests.

7. Representation of Data

Data is shared as JSON or XML, not database entities.

8. Proper HTTP Status Codes

200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Server Error

9. Versioning

Example: /api/v1/users

10. Consistent Naming

Use plural nouns like /users, /orders

7. Example REST API Flow

Request:

GET /users/10

Response:

```
{  
  "id": 10,
```

```
"name": "Meenakshi",  
"role": "BUYER"  
}
```

8. What is SOAP (Simple Explanation)

SOAP (Simple Object Access Protocol) is another way of building APIs.

Unlike REST, SOAP is a strict protocol with fixed rules and uses XML for communication.

9. How SOAP Works

SOAP uses:

- XML only
- One service endpoint
- Operations (actions) defined explicitly

Example SOAP Request:

```
<soap:Envelope>  
  <soap:Body>  
    <getUserById>  
      <id>10</id>  
    </getUserById>  
  </soap:Body>  
</soap:Envelope>
```

10. SOAP Rules

- Strict XML structure
- WSDL contract defines operations
- Strong typing
- Built-in security and transactions
- Action-based design

11. REST vs SOAP – Complete Comparison

REST:

- Lightweight
- Uses JSON/XML
- Resource-based
- Easy to use
- Fast and scalable

SOAP:

- Heavy
- XML only
- Operation-based

- Strict contracts
- Slower but secure

12. When to Use REST or SOAP

Use REST when:

- Building web/mobile APIs
- Performance matters
- Simplicity is required

Use SOAP when:

- Banking or payment systems
- High security and transactions are needed
- Legacy enterprise systems

13. Golden Summary

API = Communication bridge

REST = Rules for designing simple, scalable APIs

SOAP = Strict protocol for secure, operation-based APIs